Md Shahanur Islam Shagor
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I am writing to apply for the Software & AI Lead position at BugSense Diagnostics. Your work on paper-based diagnostic tests for urinary tract infections and antibiotic resistance is exactly the kind of real-world AI challenge I want to be part of. The opportunity to build the computer vision and machine learning stack from the ground up, with direct impact on clinical decisions, aligns with my background and interests.

Through research and development work, I have built practical computer vision systems. One example is WVAB (Wireless Vision-Aid for the Blind), a real-time assistive system I architected using YOLOv8n for object detection. The system runs fully offline on low-cost hardware, performing colour-based analysis and delivering directional audio feedback. I worked on model selection, inference optimization, and the full capture-to-result pipeline using Python, C++, and edge devices like Raspberry Pi and ESP32-CAM. This experience directly maps to the computer vision and image interpretation work at BugSense.

I am comfortable working across the hardware-software boundary. In the WVAB project, I designed a custom UDP streaming protocol to connect the camera module to the processing unit, ensuring ultra-low latency communication. I also have experience deploying software on Linux servers with Docker and Nginx, and I understand the importance of building systems that are reliable, documented, and ready for production environments.

In addition to my technical experience, I work effectively across multidisciplinary teams and communicate clearly with both technical and non-technical stakeholders. I am fluent in Bengali and English, have working knowledge of Russian and Hindi, and enjoy collaborating in international and cross-cultural environments.

I have attached my CV for your review. I would welcome the chance to discuss how my background in computer vision, AI, and software engineering can contribute to BugSense's mission.

Sincerely,

Md Shahanur Islam Shagor
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